The Radiative Sector as the Symplectic Completion of the Drift Relaxation

the conjugate momentum, propagating gravitons, and the speed of gravity (ledger D5)

Evaluation note — FRC `21-gravity`

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Abstract

The body builds the static gravitational field as a gradient flow $\dot{u} = \kappa\Delta u$, a dissipative relaxation. A referee notes correctly that a gradient flow cannot carry propagating tensor waves, and asks for the conjugate variable that converts relaxation into radiation. This note supplies it. The substrate drive is a permutation of the phase cycle, hence reversible and measure-preserving; a reversible dynamics is symplectic, so the phase offset u has a conjugate momentum π — the local winding-rate fluctuation, the quarter-turn dual of u. The gradient flow is then only the *overdamped* (static) projection; the physical dynamics is the conservative wave equation $\chi\ddot{u} = \kappa\Delta u$, whose static slice is the Newtonian field and whose propagating modes are gravitational waves. Channel unity — the same lemma that gives $\gamma = 1$ — fixes $\kappa/\chi = c^2$, so the waves travel at the speed of light. Repeated on the Fierz–Pauli field $h_{\mu\nu}$, the symplectic completion yields the hyperbolic spin-two theory with exactly two transverse-traceless polarisations of helicity ± 2, and the leading mass-quadrupole radiation recovers the binary-pulsar damping the body assumed. The transverse- traceless rank, the helicity, and the absence of doubling are exact; the long-wave speed is [approx]. Checks in `validation/validate_radiative.py`.

1 The static reading is overdamped; the substrate is reversible

The body's Kuramoto law $\dot{u}_x = \kappa(\Delta u)_x$ (a first-order, diffusion-type relaxation) is the appropriate description of *statics*: at equilibrium $\dot{u} = 0$ and the configuration solves the Poisson equation $\kappa\Delta u = -\rho$ irrespective of the time order. It cannot, however, propagate: a gradient flow has no oscillating modes. The resolution is that the substrate carries no dissipation. The drive $x \mapsto gx$ is multiplication by a generator, a permutation of the finite cycle: bijective, measure-preserving, and reversible. A reversible autonomous dynamics is symplectic, so the phase variable u is one half of a conjugate pair.

Lemma 1 (The conjugate momentum is the quarter-turn dual)**.** *The variable conjugate to the phase offset u_x is the local winding-rate fluctuation π_x — the rate at which the cell's phase advances against the drive. The drive rotates the pair (u, π) by the quarter-turn* i *of the frame ($u \mapsto \pi \mapsto -u$), exactly the position–momentum (Fourier-conjugate) rotation Q_4 of the corpus. The dissipative gradient flow is the overdamped limit in which π is adiabatically eliminated; the reversible substrate retains it.*

2 The wave equation and the speed of gravity

With the conjugate momentum restored, the discrete dynamics is generated by the Hamiltonian

$$H[u, \pi] = \sum_x \frac{\pi_x^2}{2\chi} + \frac{\kappa}{2} \sum_{\langle xy \rangle} (u_x - u_y)^2,$$

the elastic energy of the body's Lemma 4.1 plus the kinetic term reversibility forces. Hamilton's equations $\dot{u} = \pi/\chi$, $\dot{\pi} = \kappa\Delta u$ eliminate π to give

$$\chi\, \ddot{u}_x = \kappa\, (\Delta u)_x,$$

the discrete wave equation. Its static slice $\ddot{u} = 0$ is the Newtonian Poisson equation; its source-free modes propagate with dispersion $\omega^2 = (\kappa/\chi) \sum_\mu 4\sin^2(k_\mu/2)$, group velocity $\to \sqrt{\kappa/\chi}$ in the long-wave limit.

Proposition 2 (Gravity propagates at c)**.** *The kinetic coefficient χ is the temporal-correlation capacity (the cost of changing the winding rate); the stiffness κ is the spatial-correlation capacity. By channel unity (the body's Lemma pre:unity) these are one capacity read two ways, so their ratio is the space–time conversion $\kappa/\chi = c^2$. Hence the wave speed is c: gravitational waves travel at the speed of light, with the body's $\gamma = 1$ and this result two readings of the same lemma.*

3 The tensor sector: two propagating gravitons

The same completion applies to the symmetric deformation field $h_{\mu\nu}$ of the body's Proposition prop:deform, with the gauge-invariant Fierz–Pauli stiffness $E_{\mathrm{FP}}[h]$ (the body's (**??**)) as the potential and a kinetic term $\frac{1}{2\chi}\pi_{\mu\nu}\pi^{\mu\nu}$:

$$H = \sum_x \left[\tfrac{1}{2\chi}\pi_{\mu\nu}\pi^{\mu\nu}\right] + E_{\mathrm{FP}}[h] \quad \Longrightarrow \quad \chi\, \ddot{h}_{\mu\nu} = (\text{linearised Einstein operator})h_{\mu\nu}.$$

Proposition 3 (Massless spin-two, two polarisations)**.** *The linearised diffeomorphism gauge of Proposition prop:gaugeact removes four components and the four first-class constraints remove four more, leaving $10 - 4 - 4 = 2$ propagating degrees of freedom. In momentum space these are the transverse-traceless tensors: the spin-two projector $\Lambda_{ij,kl} = P_{ik}P_{jl} - \frac{1}{2}P_{ij}P_{kl}$, $P = \mathbb{1} - \hat{k}\hat{k}$, has rank exactly 2 for every propagation direction, and under a rotation by ψ about $\hat{k}$ the two modes $(h_+, h_\times)$ rotate by 2ψ — helicity ± 2. The graviton is therefore massless, spin-two, with two polarisations travelling at c.*

4 Radiation, and the residue

With the hyperbolic theory in hand, the leading radiation follows as in the linear theory: source conservation $\nabla_\mu T^{\mu\nu} = 0$ kills the monopole and dipole, and the mass quadrupole radiates with the standard luminosity, recovering the binary-pulsar damping the body's Remark rem:branch had assumed from "the shared linear theory." That assumption is now grounded. The Lense–Thirring coefficient and the preferred-frame parameters $\alpha_{1,2} = O(\Omega^{-1/2})$ likewise follow from the propagating theory rather than being posited.

Remark 4 (Noetherian provenance). The sector is Noetherian throughout. The conservative dynamics that replaces the dissipative flow is the Noether charge of *time-translation*: the drive is the same at every chronon, so a reversible autonomous evolution conserves a Hamiltonian, and a conserved Hamiltonian is symplectic — the conjugate momentum exists *because* the drive is time-homogeneous (first theorem). The reduction from ten components to two polarisations is Noether's *second* theorem on the diffeomorphism gauge — the same local symmetry whose Ward identity controls the nonlinear completion. The radiation pattern is fixed by source conservation $\nabla_\mu T^{\mu\nu} = 0$, the Noether current of cardinality and torsor equivariance. Time-translation gives the wave, gauge gives the two helicities, source conservation gives the quadrupole.

Remark 5 (No fermion-doubling). The propagating sector uses the second-difference Laplacian, symbol $4\sin^2(k/2)$, which vanishes only at $k = 0$: there is no doubling and no spurious massless graviton at the Brillouin-zone edge (where $\omega^2 = 4$ is finite). The central-difference first derivatives of the gauge functional carry the symbol $\sin k$, whose zone-edge zeros are pure gauge and are removed by the transverse- traceless projection.

Status. Ledger D5 moves from O to T | B: the *propagating* tensor sector — the conservative wave equation, two TT gravitons of helicity ± 2 at speed c, and the quadrupole radiation — is a theorem given the symplectic completion (reversibility) and the quarter-turn conjugate-momentum bridge (the corpus Q_4). The remaining residue is the fully nonlinear self-interaction; the body's Theorem thm:branch already shows the relational phase forbids tensor self-sourcing (what gravitates is winding rate), so the nonlinear completion is the exponential-metric reading with corrections in field gradient over capacity, not in graviton loops. Checks float-free in `validation/validate_radiative.py`.